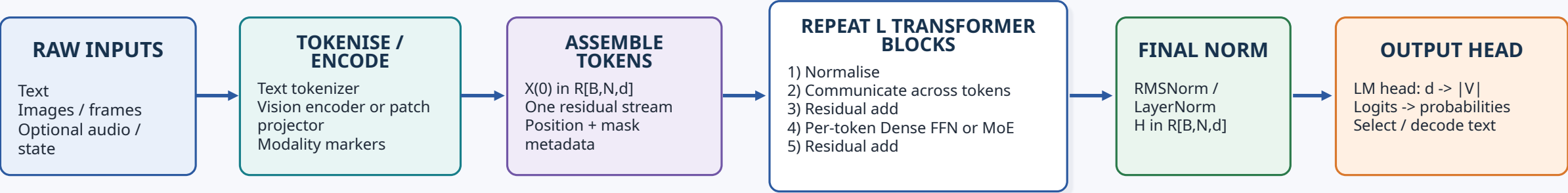


General Multimodal Transformer Architecture

16:9 / 4K-friendly reference

A widescreen, implementation-oriented symbolic map; canonical core plus a Gemma 4 26B-A4B reference mapping.



$$X(0) = \text{Assemble}(\text{Embed_text}(\text{tokens}), \text{Project_vision}(\text{VisionEncoder}(\text{image})), \text{type/position markers})$$

For layer l :
$$H(l) = X(l) + \text{Attention}(\text{Norm}_1(X(l)); \text{mask}, \text{position})$$
$$X(l+1) = H(l) + \text{Channel}(\text{Norm}_2(H(l))) \quad \text{where Channel} = \text{DenseFFN or MoE}$$

$$\text{Logits} = \text{OutputHead}(\text{FinalNorm}(X(L))) ; \quad P = \text{softmax}(\text{Logits}); \quad \text{output} = \text{Decode}(P)$$

The invariant implementation contract

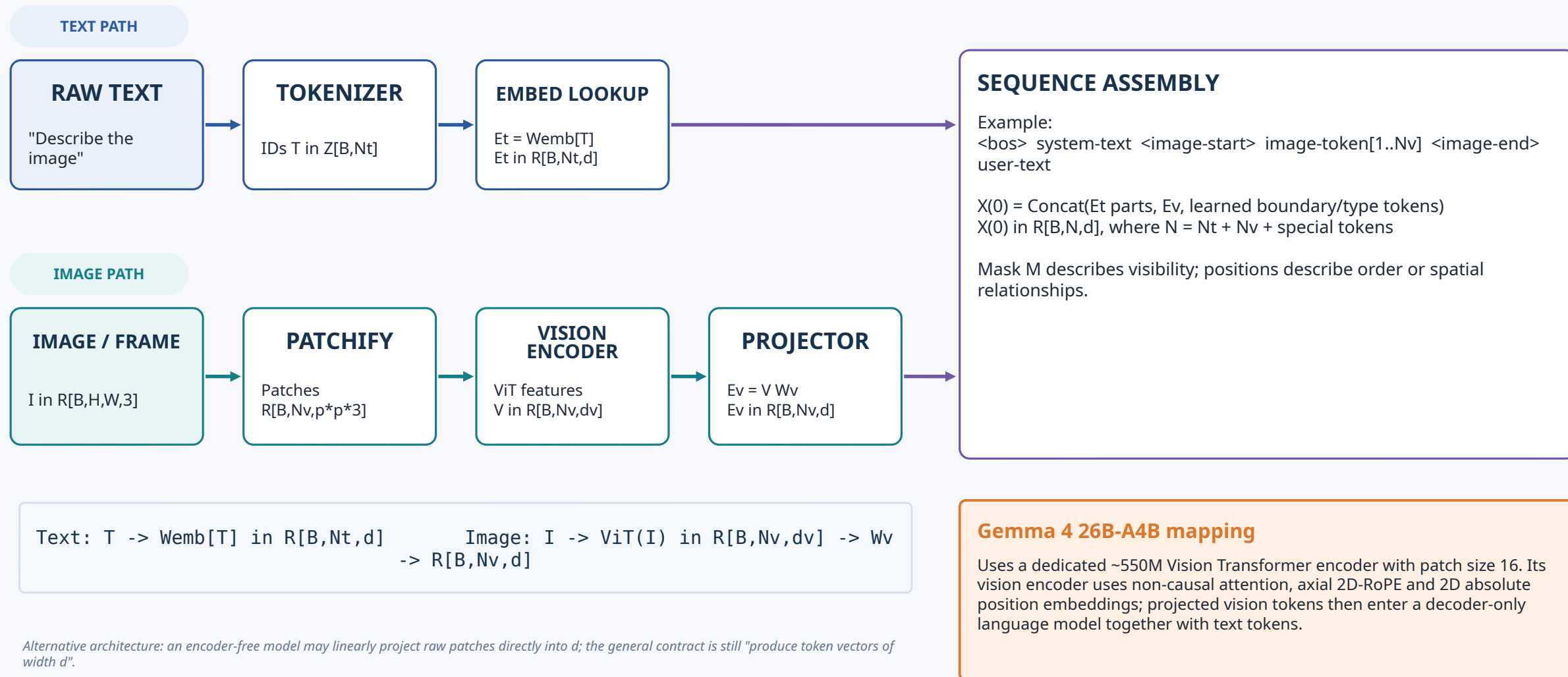
The residual stream keeps shape $[B,N,d]$. Attention mixes information across token positions. Dense FFN or MoE transforms feature channels independently for each token. Every sublayer returns $[B,N,d]$ so residual addition is valid.

Concrete reference: Gemma 4 26B-A4B

Decoder-only, text + image input, text output. 25.2B total / 3.8B active, 30 layers, 128 routed experts with 8 selected plus 1 shared, 256K context. The following pages separate the general core from Gemma-specific choices.

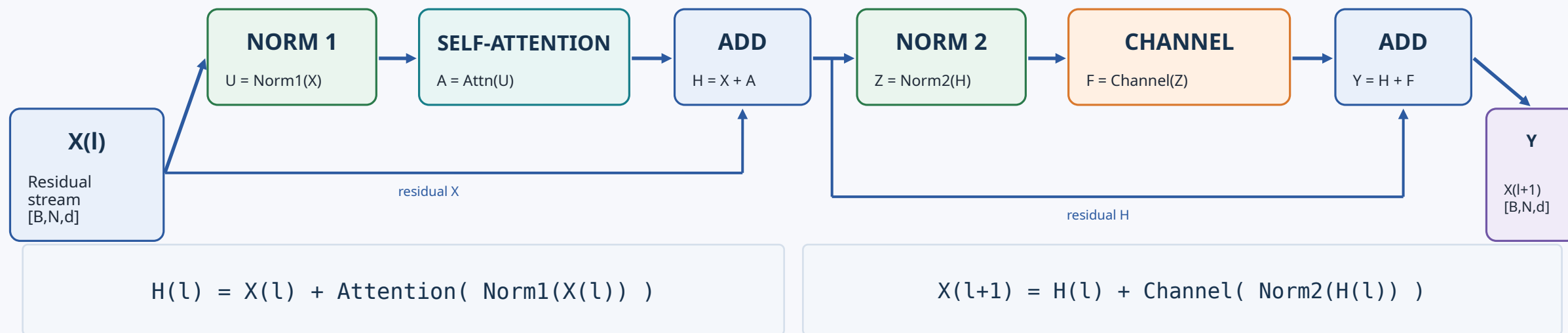
1. Multimodal input: raw data becomes one residual token stream

The Transformer consumes vectors. Domain-specific encoders and projectors must produce the common model width d .



2. One canonical pre-normalised decoder block

This is the clean baseline used throughout the implementation pseudocode. It is common, but not the only valid norm arrangement.



Why call it pre-normalised?

Because normalisation is applied before each nonlinear sublayer F:

Pre-Norm: $y = x + F(\text{Norm}(x))$

The residual path $x \rightarrow y$ stays an explicit identity path.

Post-Norm is different

The original 2017 Transformer used:

Post-Norm: $y = \text{Norm}(x + F(x))$

Here the combined residual result is normalised after the sublayer.

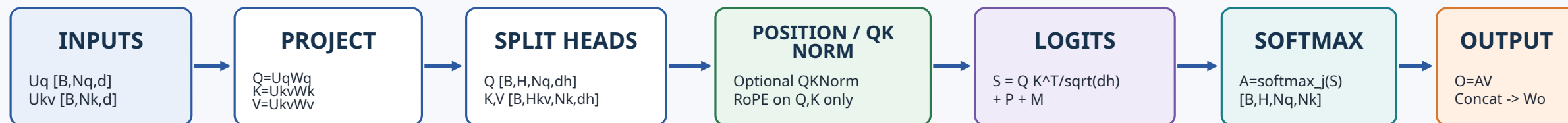
Modern models can be hybrid

Gemma 4 reports both pre- and post-RMSNorm plus QKNorm. Therefore the simple formula above is a canonical baseline, not a claim that every current model is purely Pre-Norm.

Implementation rule: choose and document one norm topology; checkpoint weights are not interchangeable between Pre-Norm, Post-Norm and hybrid blocks.

3. Attention: exact computation, shapes, position and masks

Attention produces a query-length output of width d. Softmax is always over the key index j.



$$S[h,i,j] = (\text{qhat}[h,i] \text{ dot } \text{khat}[h,j]) / \text{sqrt}(\text{dh}) + P[h,i,j] + M[i,j]$$

$M[i,j] = 0$ if token i may read token j
 $M[i,j] = -\text{infinity}$ otherwise; $\exp(-\text{infinity}) = 0$

$A[h,i,j] = \exp(S[h,i,j]) / \sum_r \exp(S[h,i,r])$
 $O[h,i,:] = \sum_j A[h,i,j] V[h,j,:]$

CAUSAL MASK

Query i may read keys j <= i.

[0	-inf	-inf	-inf]
[0	0	-inf	-inf]
[0	0	0	-inf]
[0	0	0	0]

LOCAL / SLIDING WINDOW

Allow only $|i-j| \leq w$ (plus causal rule for a decoder).

Reduces attention and KV work for long contexts; global layers restore long-range communication.

GEMMA 4 REFERENCE

26B-A4B interleaves local and global attention in a 5:1 pattern; local window = 1024. It uses RoPE locally and p-RoPE globally, and reports QKNorm. Global layers reuse keys as values.

P is optional relational/position bias. With RoPE, position is normally represented by rotating Q and K, so P may be zero. M is a logit mask, not a 0/1 multiplier.

4. Channel transform: dense gated FFN or sparse Mixture-of-Experts

This sublayer is token-wise: it does not mix sequence positions. It maps $d \rightarrow d$ so the residual add remains valid.



WHAT CHANGES?

Dense: every token executes the same FFN weights.

MoE: each token executes only a small selected subset of many expert FFNs. The router is learned and token-wise.

WHAT DOES NOT CHANGE?

The input and output width remain d .

Attention remains a separate token-mixing sublayer. MoE is not required by the Transformer definition; it is conditional computation for efficient capacity scaling.

GEMMA 4 26B-A4B

128 routed experts; 8 selected per token; 1 shared expert.

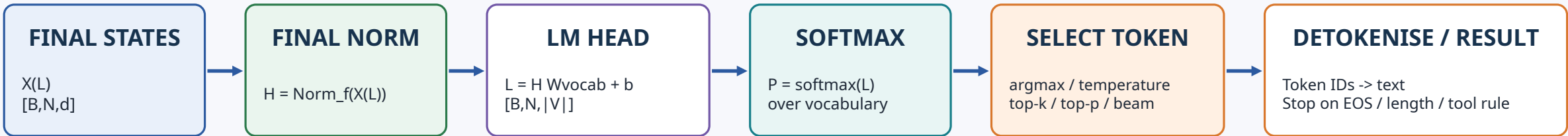
25.2B total parameters; about 3.8B active. This increases stored capacity without executing all expert parameters for every token.

Implementation hazards

Capacity limits, load balancing, token dispatch/all-to-all communication, shared experts, router precision and expert-parallel sharding are MoE system concerns; the mathematical interface still remains $\text{Channel}([B, N, d]) \rightarrow [B, N, d]$.

5. Output path: hidden states become loss during training or tokens during inference

The neural network outputs logits. Sampling, stopping and text decoding are outside the Transformer block stack.



TRAINING

SHIFTED AUTOREGRESSIVE TARGETS

Decoder input : [BOS, y1, y2, ..., y(T-1)]
Labels : [y1, y2, y3, ..., yT]

$$\text{Loss} = - \sum_t \log P(\text{label}_t \mid \text{prefix}_t, \text{multimodal context})$$

$$\text{CrossEntropy} = -\text{mean_valid} \log \text{softmax}(\text{Logits})[\text{label}]$$

INFERENCE

AUTOREGRESSIVE LOOP

1. Prefill the prompt and cache K,V for every layer.
2. For the newest token, compute one new Q,K,V per layer.
3. Attend to cached K,V, produce next-token logits, select and append.
4. Repeat until a stopping condition.

$$\text{Kcache}(l,t+1) = \text{concat}(\text{Kcache}(l,t), \text{Knew}(l,t+1)); \text{ same for } V$$

Optional speed layer: multi-token prediction (MTP)

A separate drafter predicts multiple future tokens for speculative verification. Gemma 4 releases an autoregressive MTP drafter; it is an inference accelerator around the main decoder, not a replacement for the main Transformer output head.

6. Implementation pseudocode: model forward, block and attention

Pseudocode uses the canonical Pre-Norm baseline; replace the norm topology to reproduce a specific checkpoint.

```
function encode_multimodal(text, image):
    ids = tokenizer(text)                # [B,Nt]
    text_tokens = embedding(ids)         # [B,Nt,d]

    if image exists:
        vision = vision_encoder(patchify(image)) # [B,Nv,dv]
        image_tokens = vision @ W_vision        # [B,Nv,d]
        x, positions, mask = assemble(text_tokens, image_tokens)
    else:
        x, positions, mask = assemble(text_tokens)

    return x, positions, mask            # x: [B,N,d]
```

```
function transformer_block(x, positions, mask):
    u = norm1(x)
    a = self_attention(u, positions, mask)    # [B,N,d]
    h = x + dropout(a)

    z = norm2(h)
    f = dense_ffn(z) or moe(z)               # [B,N,d]
    y = h + dropout(f)
    return y
```

```
function attention(u, positions, mask):
    q, k, v = project_qkv(u)
    q, k, v = split_heads(q, k, v)
    q, k = optional_qk_norm(q, k)
    q, k = apply_rope(q, k, positions)

    scores = q @ transpose(k) / sqrt(d_head)
    scores += optional_position_bias
    scores += additive_mask(mask)            # 0 or -infinity
    weights = softmax(scores, key_axis)

    o = weights @ v
    return merge_heads(o) @ W_output
```

```
function model_forward(text, image):
    x, pos, mask = encode_multimodal(text, image)
    for block in blocks:
        x = block(x, pos, mask)

    hidden = final_norm(x)
    logits = hidden @ W_vocab + b_vocab      # [B,N,|V|]
    return logits
```

7. Shape ledger and invariants: assertions that catch most implementation bugs

Use these as runtime assertions before optimising kernels or introducing tensor/model/expert parallelism.

Component	Symbol	Shape	Invariant / check
Token residual stream	X, U, H, Z, Y	[B, N, d]	Must keep d so residual addition is valid
Q projection	Q	[B, Hq, Nq, dh]	Hq * dh is the projected query width
K / V projection	K, V	[B, Hkv, Nk, dh]	Hkv may be smaller for GQA/MQA
Attention logits / weights	S, A	[B, Hq, Nq, Nk]	Softmax on the last/key axis Nk
Head output	Ohead	[B, Hq, Nq, dh]	Weighted sum of V
Merged attention output	O	[B, Nq, d]	Output projection returns residual width
Dense FFN hidden	Fhidden	[B, N, dff]	Expansion width; often gated
MoE router logits	R	[B, N, E]	Top-k expert selection per token
LM logits	Logits	[B, N, V]	Softmax only when probabilities are needed
KV cache per layer	Kcache, Vcache	[B, Hkv, T, dh]	T grows during deep decoding (prompt+unfilled ones)

Critical semantic checks

RoPE modifies Q and K, not V. A logit mask is 0/-infinity before softmax. Mist: rounding is token-wise. Attention output length is Hq * Nq length is Nk. Padding tokens must be excluded from loss and attention.

8. Three Transformer system topologies

The same attention + channel block can be arranged as an encoder, a causal decoder, or an encoder-decoder system.

ENCODER-ONLY

Input tokens

- > bidirectional self-attention blocks
- > final norm
- > pooled or per-token task head

Mask: all valid tokens may communicate.

Typical outputs: class label, embedding, segmentation, token tags.

No autoregressive token loop is required.

understanding / embedding

DECODER-ONLY

Prompt and optional multimodal tokens

- > causal self-attention blocks
- > final norm
- > LM head
- > next-token generation loop

Mask: token i may read $j \leq i$.

Gemma 4 26B-A4B belongs here, with a separate image encoder feeding visual tokens into the prompt sequence.

generation / agents

ENCODER-DECODER

Source input

- > bidirectional encoder -> memory C

Target prefix

- > causal self-attention
- > cross-attention(Q from decoder; K,V from C)
- > channel transform -> LM head

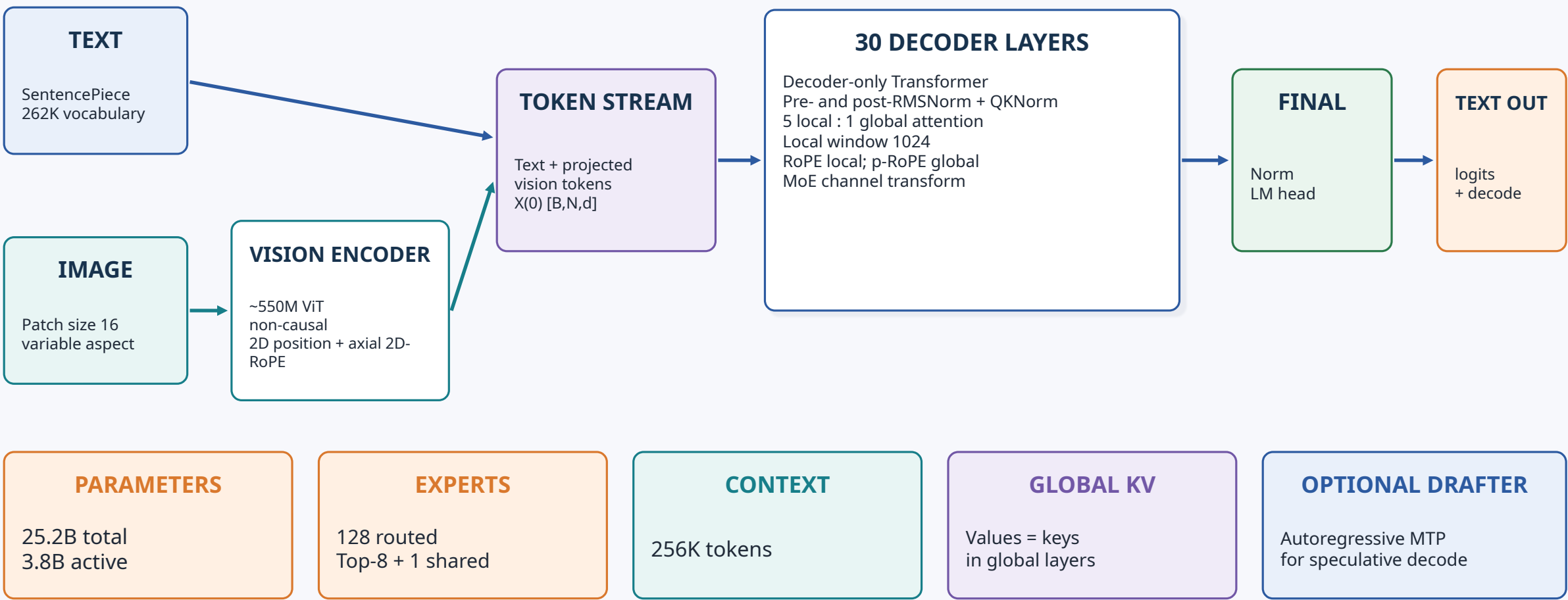
Useful when source and target have separate structures.

translation / seq2seq

Shared core: encode entities -> [token communication + token-wise nonlinear transform]^L -> task-specific output head

9. Concrete reference map: Gemma 4 26B-A4B

Official high-level model facts mapped onto the general symbolic architecture.



Important: the official report/model card are high-level. For binary-compatible implementation, use the released model config and reference code as the authority for exact tensor ordering, norm placement, parameter sharing and checkpoint names.

10. Pre-Norm, Post-Norm and hybrid normalisation: the precise answer

"Pre-normalised Transformer" describes where Norm is placed relative to Attention/FFN, not a different attention law.

PRE-NORM

```
u = Norm(x)
y = x + F(u)
```

Norm occurs before the sublayer.
The identity residual path is explicit.

This was the baseline in the previous architecture because it gives a clean, common implementation contract and has historically improved optimisation stability for deep stacks.

POST-NORM

```
u = F(x)
y = Norm(x + u)
```

Norm occurs after residual addition.

The original Attention Is All You Need encoder/decoder used Post-Norm around each sublayer.

HYBRID / PERI-NORM

A modern block may normalise before and after sublayers, normalise Q and K, or use different rules for attention and MoE.

Gemma 4 explicitly reports pre-norm and post-norm with RMSNorm, plus QKNorm. Therefore an exact Gemma implementation is not the pure Pre-Norm baseline.

The previous formula was called Pre-Norm because it literally used `Attention(Norm(X))` and `Channel(Norm(H))` before each residual add.

Practical advice

Implement the general interfaces first, but parameterise the norm topology. To load real weights, reproduce the reference model exactly rather than assuming that all modern Transformers use one universal normalisation placement.

References and implementation notes

Primary sources used for the real-model mapping and the normalisation terminology.

1. Gemma 4 model card - Google AI for Developers

Official model overview and table: modalities, dense/MoE variants, 26B-A4B parameter counts, layers, expert count, context, vision encoder, local/global attention overview.
https://ai.google.dev/gemma/docs/core/model_card_4

2. Gemma 4 Technical Report - Google DeepMind, arXiv:2607.02770

Decoder-only architecture; pre- and post-RMSNorm; QKNorm; local/global ratio; p-RoPE; vision encoder details; parameters; MTP drafter.
<https://arxiv.org/abs/2607.02770>

3. Attention Is All You Need - Vaswani et al., 2017

Original encoder-decoder Transformer, scaled dot-product attention, multi-head attention, residual connections, Post-Norm and position-wise feed-forward networks.
<https://arxiv.org/abs/1706.03762>

4. On Layer Normalization in the Transformer Architecture - Xiong et al., 2020

Formal comparison of Post-LN and Pre-LN placement and their training behaviour.
<https://arxiv.org/abs/2002.04745>

Document scope

The symbolic architecture is framework-agnostic and implementation-oriented. It intentionally separates invariant interfaces from model-specific choices such as norm topology, GQA/MQA, hybrid attention, expert routing, p-RoPE, MTP and vision front-ends.